24-HOUR HACKATHON PROJECT

FusionShield AI

Multi-Layer AI-Driven Cyber Defense

Detecting Coordinated Scam Attacks Across Text & Voice Channels

Email | SMS | Chat

Voice Intelligence

Cross-Channel Correlation

PROBLEM STATEMENT

The Evolving Cyber Threat Landscape

Email Phishing

Deceptive messages designed to steal credentials and financial information

SMS & Chat

Smishing and chat-based lures targeting mobile users

Voice Calls

Vishing using synthetic or scripted voices to manipulate victims

The Critical Gap in Existing Solutions

Individually, these signals may appear harmless. **Together, they form a coordinated scam.**

Existing solutions focus on single-channel detection and **fail to identify cross-channel relationships.**

SOLUTION

Introducing FusionShield AI

A multi-layer AI pipeline that integrates text and voice intelligence, combines signals through a fusion engine, and detects coordinated cross-channel attacks.

Text Intelligence

NLP + DistilBERT analyzes email, SMS, and chat for phishing patterns, urgency signals, and financial intent.

Voice Intelligence

Resemblyzer extracts speaker embeddings and signal features to detect anomalies and synthetic speech.

Fusion Engine

Weighted combination with correlation amplification transforms weak individual signals into a strong threat signal.

Correlation Engine

Detects cross-channel relationships and shared intent across communication channels — the key innovation.

SYSTEM ARCHITECTURE

4-Layer Intelligent Pipeline

Email / SMS / Chat

Audio Input (Optional)

Layer 01

Text Intelligence Layer

DistilBERT · NLP · Intent Detection

Layer 02

Voice Intelligence Layer

Resemblyzer · Signal Feature Extraction

Layer 03

Fusion Engine

Weighted Fusion · Correlation Amplification

Layer 04

Correlation Engine

Cross-Channel Pattern Detection

OUTPUT · Explainability & Visualization Layer

- Threat Level
- Confidence Score
- AI Reasoning
- Attack Timeline
- Gauge Chart
- Fusion Graph

KEY INNOVATION

Correlation Engine

FusionShield detects relationships between channels by checking if inputs share similar intent (banking, urgency, OTP) and identifying coordinated attack patterns across platforms.

Intent Matching

- Analyzes text and voice inputs for semantic similarity
- Detects shared keywords: OTP, bank, verification, urgent

Cross-Channel Linking

- Connects signals that arrive on different channels
- Identifies timing patterns and sequencing of attacks

Correlation Amplification

- Boosts threat score when multiple channels align
- Transforms weak individual signals into strong threat signal

Result: **Coordinated Cross-Channel Attack Detected — with full explainability and confidence scoring**

TECH STACK

Technologies & Components

Frontend

React
UI development & component architecture

Tailwind CSS
Utility-first responsive styling

Recharts
Gauge charts & fusion graphs

Backend

FastAPI
High-performance REST API server

Uvicorn
ASGI runtime for backend services

AI Components

DistilBERT
NLP text analysis & intent detection

Resemblyzer
Voice embeddings & speaker analysis

Keyword Intent
Semantic keyword-based understanding

Signal Features
Voice signal extraction & analysis

FEATURES & USE CASES

Capabilities & Real-World Applications

Core Features

- Multi-channel detection — text and voice
- Confidence scoring per analysis
- Multi-platform text support (email, SMS, chat)
- Explainable AI outputs with reasoning
- Cross-channel correlation detection
- Attack timeline simulation
- Weighted fusion engine with amplification
- Gauge and fusion graph visualization

Use Cases

Individuals

Verifying suspicious messages or calls before acting on them

Banking & Fintech

Real-time fraud detection across all communication channels

Enterprise Security

Email and communication security for large organizations

IMPACT

Why FusionShield Matters

AI-Driven Scam Detection

Detects modern, AI-powered scams that evade traditional single-channel filters

Financial Fraud Prevention

Prevents financial loss by identifying coordinated attacks before victims respond

Multi-Platform Security

Improves security posture across email, SMS, chat, and voice simultaneously

New Defense Paradigm

Introduces a multi-channel cyber defense approach not found in existing solutions

FusionShield AI — Multi-channel cyber defense, powered by AI, designed for the modern threat landscape.